Rohan Pandey

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EDUCATION

University of Washington, Seattle, WA — B.S. March 2026 • M.S. Fall 2026 –

B.S. in Applied & Computational Mathematical Sciences — Scientific Computing and Numerical Analysis • Returning Fall 2026 for M.S. in ACMS

Minor: Neural Computation & Engineering

Relevant Coursework: Machine / Reinforcement Learning, Data Structures & Algorithms, High-Performance Scientific Computing, Optimization (Linear / Nonlinear / Discrete), Database Systems, Advanced Neurotechnology, Data Science

Awards: NASA Space Grant Awardee

EXPERIENCE

Software Engineer II, DigitalOcean — Dedicated Inference Team — March 2026 – Present

- Build benchmarking infrastructure on the **Dedicated Inference** team — orchestrate **vLLM** on **Kubernetes** to evaluate GPU / inference throughput and latency across heterogeneous hardware and model configurations; results directly inform capacity, routing, and SLA decisions for production inference workloads
- Own production observability for the inference fleet: **Prometheus** metrics, **Grafana** dashboards, and **PagerDuty** alerting; perform **GPU computing and performance debugging** across the Linux + networking stack to isolate throughput / latency regressions
- Designing a **Kernel Optimization AI Agent** that profiles, edits, and tunes GPU kernels via LLM tool-use, automated benchmarking, and structured search — targeting measurable speedups on real inference paths

Lead Reinforcement Learning Engineer, UW Math AI Lab — September 2025 – Present

- Designed a **PPO + Graph Neural Network** reinforcement-learning framework to synthesize efficient arithmetic circuits for polynomials; achieved **~70% success on degree-3 polynomials** via curriculum learning, symbolic verification, and shaped reward functions
- Scaled to higher-degree synthesis with **Monte Carlo Tree Search** + AlphaZero-style training and compared against SAC; co-first-authored *CircuitBuilder* at **ICLR 2026 (Rio de Janeiro)**, with co-authored follow-up work on LLM scientific reasoning also at ICLR 2026
- Released open-source benchmark code and led the RL sub-team; co-presented results to faculty and the broader Math AI Lab community

Machine Learning Engineer, Mercor — December 2025 – March 2026

- Built and deployed an end-to-end **ML pipeline** for large-scale text sentiment analysis; fine-tuned and evaluated transformer models on domain-specific data with custom scoring metrics
- Owned data preprocessing, model validation, and inference automation to support reliable experimentation and production-grade workflows

Cloud & Digital SAP Intern, PricewaterhouseCoopers — June – August 2025

- Engineered high-throughput **Python data pipelines** using vectorization and parallel I/O to reduce ingestion latency on large-scale workflows
- Designed and shipped a secure, self-service web app enabling cross-functional teams to configure, trigger, and monitor automated workflows; implemented modular components with logging and fault-handling for maintainability and scale

SELECTED PUBLICATIONS

- *CircuitBuilder: From Polynomials to Circuits via Reinforcement Learning* • **ICLR 2026** (co-first author)
- *Beyond the Answer: Decoding the Behavior of LLMs as Scientific Reasoners* • **ICLR 2026** — **R. Pandey** et al.
- *Quantization Blindspots: How Model Compression Breaks Backdoor Defenses* • arXiv preprint — **R. Pandey**, E. Ye • benchmark code

SELECTED PROJECTS & AWARDS

- **Hackathon wins:** Winner — *Husky Maps* (UW Claude Hackathon 2025); 2nd Place — *BloomSync AI* (DubHacks); Top Project — *NeuralTrace* (NeuroAI Hackathon, real-time EEG visualizer)

SKILLS

Languages: Python, C++, Java, Lean, SQL, Bash, MATLAB

ML / Inference: vLLM, PyTorch, TensorFlow, CUDA, GPU computing, Transformers, RL (PPO, DDPG, MCTS, AlphaZero), GNNs

Systems & Infra: Linux, Kubernetes, Docker, Prometheus, Grafana, PagerDuty, FastAPI, PostgreSQL, networking, performance debugging, Git/GitHub Actions, DigitalOcean

Math: Optimization (LP / NLP / IP), Numerical Methods, ODE systems, Probability, Linear Algebra